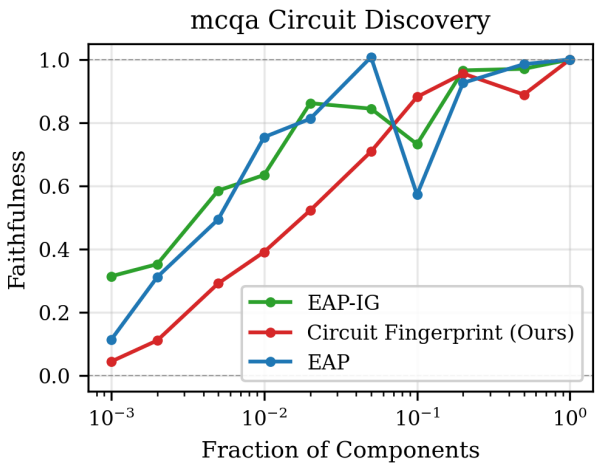
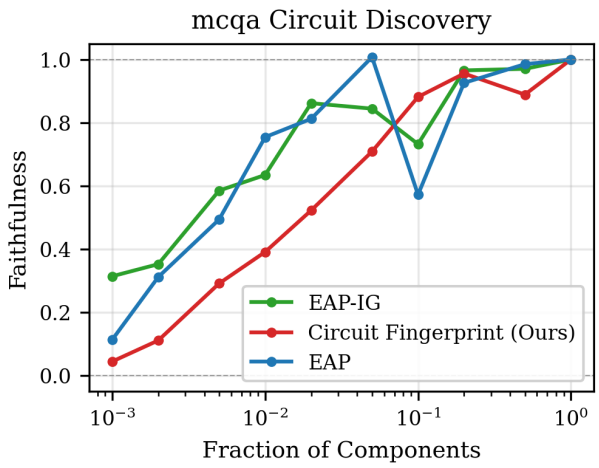
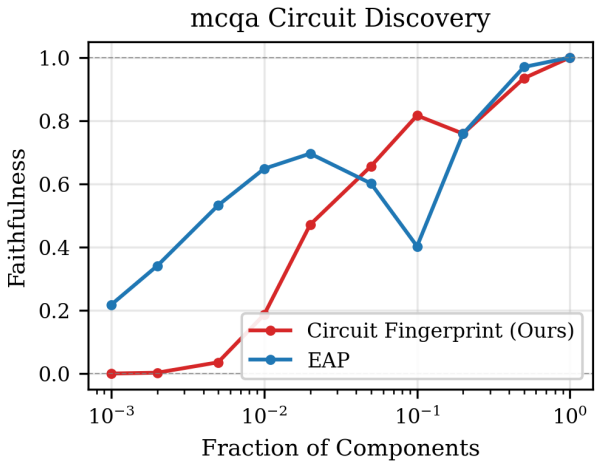


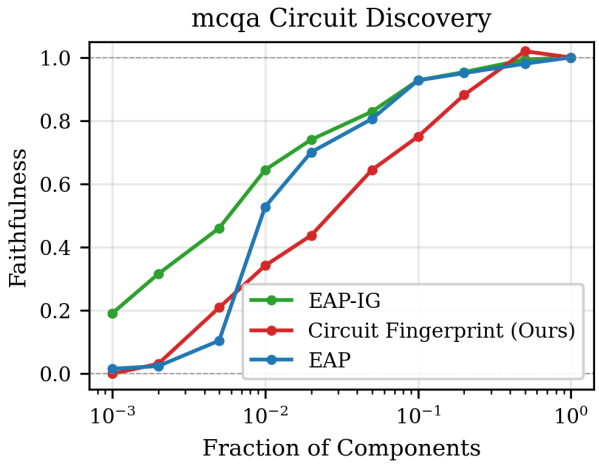
(a) GPT2-Small



(b) Qwen2.5-0.5B



(c) Llama3.2-1B

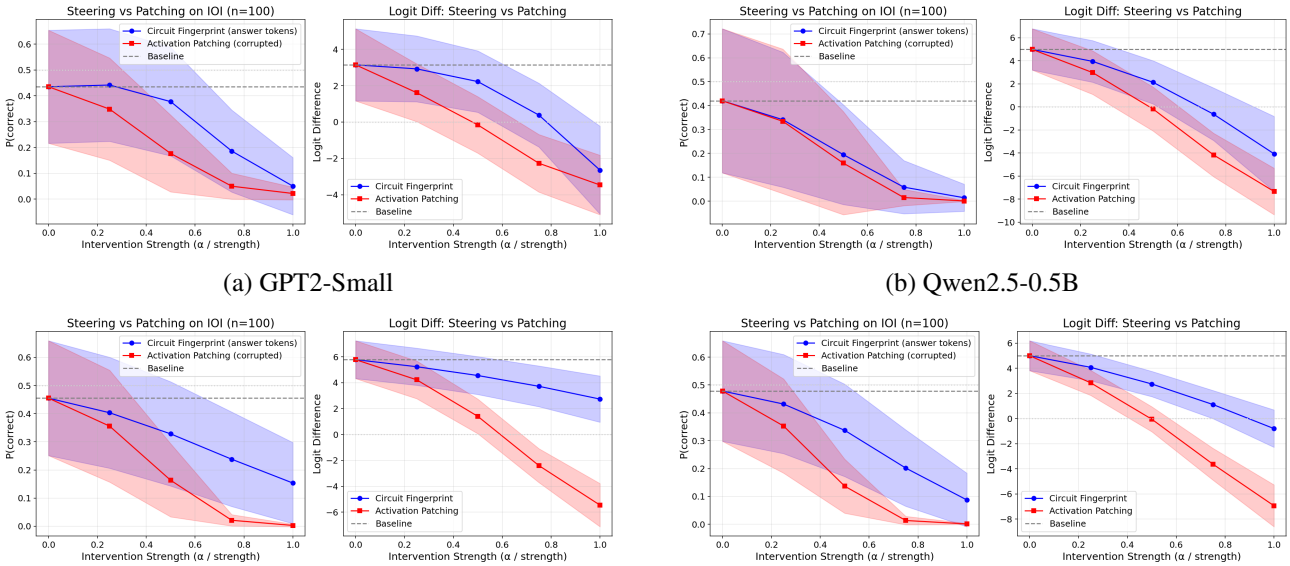


(d) OPT-1.3B

Figure 9. Edge-level circuit discovery for MCQA across models.

B. Activation Patching Comparisson

We present the complete steering validation curves for SVA and MCQA, complementing the IOI results in Figure 5. Both tasks exhibit similar patterns: answer token directions achieve comparable endpoint performance to activation patching despite using only geometric information.



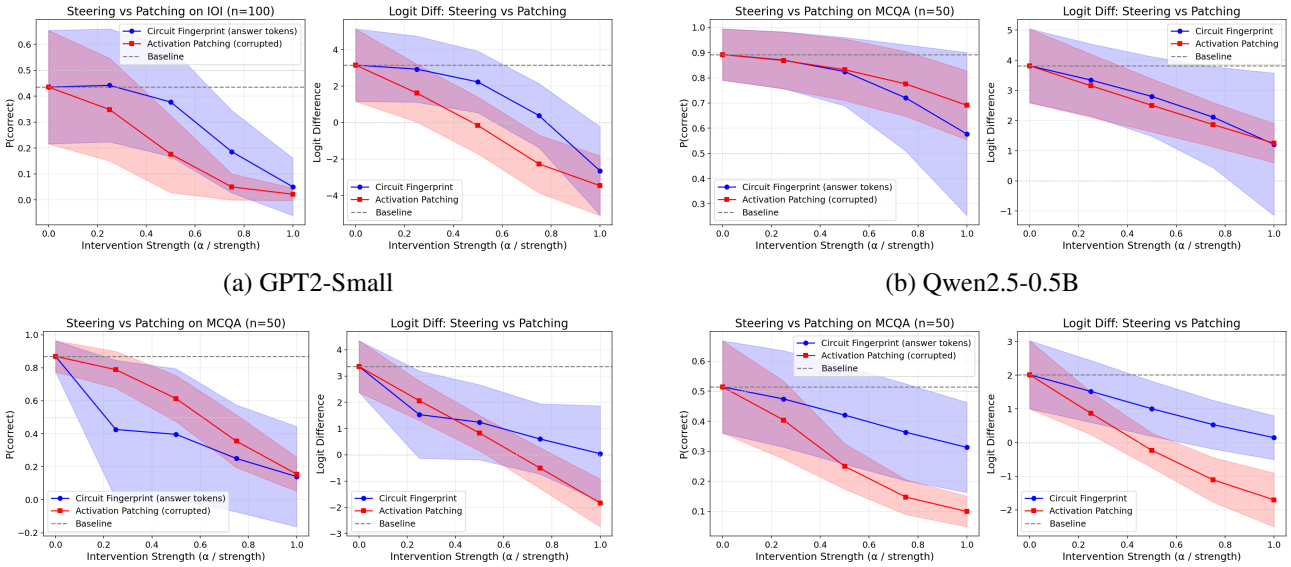
(a) GPT2-Small

(b) Qwen2.5-0.5B

(c) Llama3.2-1B

(d) OPT-1.3B

Figure 10. Activation Patching Comparisson for IOI.



(a) GPT2-Small (b) Qwen2.5-0.5B

(c) Llama3.2-1B (d) OPT-1.3B

Figure 11. Activation Patching Comparisson for MCQA.

C. Steering Class Vectors

Table 4 presents per-emotion results. Geometric steering achieves higher classification accuracy on 3/5 emotions (joy, sadness, surprise), while consistently producing lower perplexity—indicating more fluent generations—without degrading factual accuracy (89.6% vs 90.1%).

Table 4. Per-Language Emotion Steering: Geometric Method vs Instruction Prompting

Emotion	Classification Acc.		Perplexity↓		Factual Acc.
	Baseline	Steered	Baseline	Steered	Steered
Anger	55.1%	24.5%	10.50	11.12	96%
Disgust	75.5%	67.3%	17.95	13.14	78%
Joy	49.0%	**87.8%**	15.76	12.93	100%
Sadness	53.1%	**89.8%**	18.09	13.95	81%
Surprise	32.7%	**79.6%**	22.86	15.72	93%
Average	53.1%	**69.8%**	17.03	**13.37**	90%